

Problem

A three-man jury consists of two competent jurors, each making the correct decision independently with probability p , and a third juror who decides every case by flipping a fair coin. The jury follows majority rule. Compare its probability of making the correct decision with that of a one-man jury whose probability of being correct is also p .

Approach

The three-man jury reaches the correct decision in two situations:

- Both competent jurors are correct.
- The competent jurors disagree, and the coin-flipping juror happens to choose the correct verdict.

Compute this probability and compare it with the one-man jury.

Solution

The probability that both competent jurors are correct is

$$P(\text{both correct}) = p^2.$$

The probability that they disagree is

$$P(\text{disagree}) = p(1 - p) + (1 - p)p = 2p(1 - p).$$

When they disagree, the coin-flipping juror is correct with probability

$$\frac{1}{2}.$$

Hence,

$$\begin{aligned} P(\text{3-man}) &= p^2 + 2p(1 - p) \left(\frac{1}{2} \right) \\ &= p^2 + p(1 - p) \\ &= p. \end{aligned}$$

The one-man jury is correct with probability

$$P(\text{1-man}) = p.$$

Therefore,

$$P(\text{3-man}) = P(\text{1-man}),$$

so both juries are equally likely to reach the correct decision.

Key Insight

The coin-flipping juror only decides cases where the two competent jurors disagree. Since the coin is equally likely to be correct or incorrect, it adds no net accuracy, leaving the overall probability unchanged.